

HCSM-46

Graviton Exact Masslessness

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstannp@yahoo.com

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Abstract

The Hodge Complex Standard Model identifies the graviton as the self paired $B_1 \oplus B_2$ curvature mode of the symmetric tensor square $\text{Sym}^2(\Omega^1|_{\lambda=4})$ on the periodic 8×8 torus (HCSM-45). This paper proves that the graviton is exactly massless at all orders, with no finite- N , one-loop, or non perturbative correction. The proof is a Schur-orthogonality argument: any mass operator on the $\lambda = 4$ sector that couples through the trace ϵT_μ^μ (with $\epsilon = D - 4$ a Lorentz scalar) transforms in the A_1 isotypic representation of D_4 , while the graviton transforms in $B_1 \oplus B_2$; the two irreps are distinct, so the mass matrix element vanishes identically. This is the discrete analogue of the continuum statement that diffeomorphism invariance forbids a graviton mass. The paper also retracts two results from the earlier WIN Gravity Theorem: the finite- N graviton mass estimate (an imported effective-field-theory ansatz, whose own arithmetic also exceeds the LIGO bound by 13 orders of magnitude) and the Kaluza-Klein interpretation of the first excited graviton level at 1.52 TeV (a category error, since the lattice is not a compactified spatial dimension). All numerical claims are verified at machine precision.

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1 Introduction

1.1 The graviton mass question

The Hodge Complex Standard Model (HCSM) identifies the graviton as the self-paired $B_1 \oplus B_2$ curvature mode of the symmetric tensor square $\text{Sym}^2(\Omega^1|_{\lambda=4})$ on the periodic 8×8 torus (HCSM-45 [7]). The graviton sector has dimension 108 and decomposes as $54B_1 \oplus 54B_2$; its polarization space is $V = B_1 \oplus B_2$, of dimension 2.

The question addressed by this paper is whether the graviton is exactly massless, or whether it acquires a mass from the substrate's dynamics. In continuum general relativity, the graviton mass is forbidden by diffeomorphism invariance. In the HCSM substrate, the analogue of diffeomorphism invariance is the D_4 point-group symmetry of the lattice; the question is whether this symmetry forbids a graviton mass exactly.

The answer is yes. The proof is a Schur-orthogonality argument on the D_4 isotypic decomposition: any mass operator on the $\lambda = 4$ sector that couples through the trace ϵT^μ_μ transforms as A_1 , while the graviton transforms as $B_1 \oplus B_2$; the mass matrix element vanishes identically.

1.2 The one line summary

The graviton is exactly massless at all orders because its D_4 isotypic sector $B_1 \oplus B_2$ is orthogonal to the A_1 sector that hosts any trace coupled mass operator. No perturbative or non perturbative correction generates a graviton mass.

1.3 Main results

- **Structure of the mass operator (Section 3).** Any mass operator on the $\lambda = 4$ sector that couples through the trace ϵT_μ^μ transforms in the A_1 isotypic representation of D_4 .
- **Exact masslessness (Theorem 4).** The graviton is exactly massless: $m_{\text{graviton}} = 0$ at all orders.
- **Absence of finite- N corrections (Section 4).** No finite- N correction to the graviton mass exists in the current HCSM architecture. The SYK/QIN layer that motivated such a correction was removed in HCSM-00 §9.
- **Retraction of the WIN-era estimates (Section 5).** The finite- N graviton mass estimate and the Kaluza-Klein resonance interpretation of the WIN Gravity Theorem are retracted as non HCSM-native.
- **Falsifiable predictions (Section 7).** Exact masslessness, no fifth force, no scalar graviton, no Kaluza-Klein tower.

1.4 Cross references

The paper depends on the following HCSM companions:

- HCSM-00 [1]: foundations, notation, mapping; §9 retired layers.
- HCSM-02 [2]: the Hodge complex on Λ ; the $\lambda = 4$ isotypic decomposition.
- HCSM-05 [4]: orbit decomposition; the D_4 character table.
- HCSM-06 [5]: sector-coarse correspondence; the D_4 tensor-product rules.
- HCSM-45 [7]: graviton as self-paired curvature mode; the $54B_1 \oplus 54B_2$ sector.
- HCSM-47 [8]: graviton self-coupling from D_4 (downstream).
- HCSM-48 [9]: Faddeev-Popov G_N normalization (downstream).
- HCSM-49 [10]: Newton constant 8π factor (downstream).
- HCSM-50 [11]: composite-operator effective action (downstream).

1.5 Organization

Section 2 states the setup: the graviton sector and the D_4 isotypic decomposition. Section 3 proves the structure of the mass operator. Section 4 proves the Graviton Exact Masslessness Theorem and excludes finite- N corrections. Section 5 retracts the two WIN-era estimates. Section 6 states consequences for gravitational phenomenology. Section 7 gives reproducibility notes. Section 8 concludes.

2 Setup

2.1 The graviton sector

By HCSM-45 [7], the graviton is the symmetrized self-paired bilinear $h = \omega \otimes \omega$ in the substrate 1-form $\omega \in \Omega^1$, restricted to the $\lambda = 4$ sector and projected onto the $B_1 \oplus B_2$ isotypic piece of the symmetric tensor square $\text{Sym}^2(\Omega^1|_{\lambda=4})$. The symmetric square decomposes as

$$\text{Sym}^2(\Omega^1|_{\lambda=4}) = 60A_1 \oplus 46A_2 \oplus 54B_1 \oplus 54B_2 \oplus 96E, \quad (1)$$

of total dimension 406. The graviton sector is

$$V_{\text{grav}} = 54B_1 \oplus 54B_2, \quad \dim V_{\text{grav}} = 108. \quad (2)$$

Its polarization space is $V = B_1 \oplus B_2$, of dimension 2, with D_4 character $(2, 2, -2, 0, 0)$ on the five conjugacy classes.

2.2 The D_4 character table and irreps

The D_4 point group has five irreps: two 1-dimensional real irreps A_1, A_2 ; two 1-dimensional real irreps B_1, B_2 ; and one 2-dimensional real irrep E . The character table on conjugacy classes $\{e\}, \{r^2\}, \{r, r^3\}, \{s, r^2s\}, \{rs, r^3s\}$ with class sizes $(1, 1, 2, 2, 2)$ is

irrep	e	r^2	r	s	rs
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

The five irreps are pairwise non-isomorphic. In particular, A_1 and $B_1 \oplus B_2$ are distinct.

2.3 The isotypic decomposition of the $\lambda = 4$ sector

The full 56-dimensional Hodge sector at $\lambda = 4$ carries the D_4 content

$$\Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E \quad (3)$$

(HCSM-02 [2], HCSM-05 [4]). Any D_4 -invariant operator on this space decomposes block-diagonally with respect to the isotypic decomposition; by Schur's lemma, it cannot mix distinct isotypic sectors.

3 Structure of the mass operator

3.1 The trace coupling

Proposition 1 (Mass operator structure). *Any mass operator on the $\lambda = 4$ sector that couples through the trace ϵT_μ^μ (with $\epsilon = D - 4$ a Lorentz scalar and T_μ^μ the trace of the stress energy tensor) transforms in the A_1 isotypic representation of D_4 .*

Proof. The Lorentz scalar $\epsilon = D-4$ is invariant under D_4 ; it transforms in the trivial representation A_1 . The trace T_μ^μ of the stress energy tensor is also a Lorentz scalar and therefore also invariant under D_4 ; it transforms in A_1 . Their product ϵT_μ^μ transforms as the tensor product $A_1 \otimes A_1 = A_1$.

For a metric perturbation $h_{\mu\nu}$ around the flat torus, the trace h_μ^μ is the Lorentz-scalar part of $h_{\mu\nu}$; it transforms in A_1 (see, e.g., the standard decomposition of a symmetric 2-tensor into trace, traceless-symmetric, and antisymmetric parts). Any mass term that couples the metric perturbation to the trace ϵT_μ^μ is therefore of the form

$$m^2(h_\mu^\mu)^2 + \dots, \quad (4)$$

where all terms in the ellipsis transform in A_1 by the same argument.

Hence the mass operator M lies in the A_1 isotypic sector of $\text{Sym}^2(\Omega^1|_{\lambda=4})$. $\square$

Remark 2. The statement is structural: it does not depend on which specific microscopic mechanism generates the coupling ϵT_μ^μ . As long as the coupling is a Lorentz scalar times the trace of the stress-energy tensor, the mass operator transforms in A_1 .

3.2 The graviton sector

Proposition 3 (Graviton sector). *The graviton sector is $V_{\text{grav}} = 54B_1 \oplus 54B_2$, which lies entirely in the $B_1 \oplus B_2$ isotypic sector of $\text{Sym}^2(\Omega^1|_{\lambda=4})$.*

Proof. Direct from HCSM-45 [7] Theorem 3.4 and Corollary 4.2: the B_1 and B_2 multiplicities in $\text{Sym}^2(\Omega^1|_{\lambda=4})$ are both 54. $\square$

4 Exact masslessness of the graviton

4.1 Schur orthogonality

Theorem 4 (Graviton Exact Masslessness). *The graviton is exactly massless at all orders in perturbation theory and at all orders in $1/N$:*

$$m_{\text{graviton}} = 0. \quad (5)$$

Proof. By Proposition 1, the mass operator M lies in the A_1 isotypic sector of D_4 . By Proposition 3, the graviton lies in the $B_1 \oplus B_2$ isotypic sector. The irreps A_1 and B_1, B_2 are distinct irreducible representations of D_4 (Section 2.2).

By Schur's lemma, any D_4 -invariant operator is block-diagonal with respect to the isotypic decomposition. Hence

$$\langle A_1, B_1 \oplus B_2 \rangle_{D_4} = 0. \quad (6)$$

The mass operator therefore annihilates every mode in $B_1 \oplus B_2$, and the graviton mass vanishes.

The argument is exact: it does not rely on smallness of any coupling, on a perturbative expansion, or on a finite- N suppression. Since the D_4 symmetry is an exact symmetry of the discrete torus, the Schur orthogonality is exact at all orders. $\square$

Remark 5 (Discrete analogue of diffeomorphism invariance). In continuum general relativity, the graviton mass is forbidden by diffeomorphism invariance, in the same way the photon mass is forbidden by gauge invariance. On the discrete torus, the analogue of diffeomorphism invariance is the D_4 symmetry, which is an exact symmetry of the lattice. Schur orthogonality of the A_1 and $B_1 \oplus B_2$ sectors is the discrete counterpart of the statement that a mass term $m^2 h_{\mu\nu} h^{\mu\nu}$ is forbidden by diffeomorphism invariance.

4.2 Absence of finite- N corrections

Proposition 6 (No finite- N correction). *No finite- N correction to the graviton mass exists in the current HCSM architecture.*

Proof. The earlier WIN-era Gravity Theorem proposed a finite- N correction based on the SYK kernel. The SYK/QIN layer was removed from the HCSM architecture in HCSM-00 [1] §9: direct computation of the infrared structure of the Ω^1 fermionic sector establishes that site-local D_4 -invariant quartic interactions do not flow to SYK, because momentum conservation at each vertex prevents melonic routing. The correct infrared description of Ω^1 is a $(2+1)$ -dimensional local fermionic theory, not SYK.

There is no SYK kernel in the HCSM architecture. Any correction based on the SYK kernel is therefore not defined within HCSM. The remaining interactions (the D_4 -invariant quartic ϕ^4 / ω^4 terms of HCSM-18 and the $\Omega^1 / \Omega^0 / \Omega^2$ interactions) are D_4 -invariant; by the same Schur orthogonality argument as Theorem 4, none of them can generate a mass term in $A_1 \otimes (B_1 \oplus B_2) = (A_1 \otimes B_1) \oplus (A_1 \otimes B_2) = B_1 \oplus B_2 \neq A_1$. $\square$

Remark 7 (No generic quantum correction). In a generic effective field theory, quantum corrections generate masses of order the cutoff, suppressed by powers of the cutoff scale. The HCSM framework is not a generic EFT: it is a finite theory on a finite lattice with an exact D_4 symmetry. There is no continuum limit to take, no UV divergences to cancel, and no anomalous dimensions to run. The symmetry is exact, and it protects the graviton mass exactly.

5 Retractions from the WIN Gravity Theorem

5.1 Retraction of the finite- N graviton mass estimate

The WIN-era Gravity Theorem proposed a finite- N correction of the form

$$m_{\text{graviton}}^2 \lesssim \delta_{\text{edge}}^2 M_P^2 e^{-2kL} \sim 10^{-74} M_P^2. \quad (7)$$

This estimate is retracted for three independent reasons:

1. **Arithmetic error.** With $\delta_{\text{edge}} = 0.012143$ and $e^{-2kL} = 4.067 \times 10^{-34}$, the formula gives

$$\delta_{\text{edge}}^2 \cdot e^{-2kL} = 6.0 \times 10^{-38}, \quad (8)$$

not 10^{-74} . The paper’s stated value disagrees with its own formula by a factor of 10^{36} .

2. **Observational exclusion.** Even at face value, the claimed $10^{-74} M_P^2$ corresponds to $m_{\text{graviton}} \sim 10^{-9}$ eV, which exceeds the LIGO/Virgo bound $m_{\text{graviton}} < 1.2 \times 10^{-22}$ eV [12] by 13 orders of magnitude.
3. **Structural error.** The formula is a scalar ansatz: it would give mass to every D_4 irrep equally, including the A_1 trace mode, which is the mass sector itself. This contradicts the block-diagonal structure of D_4 -invariant operators (Schur’s lemma).

The estimate is replaced by Theorem 4.

5.2 Retraction of the Kaluza-Klein resonance interpretation

The WIN-era Gravity Theorem interpreted the first excited graviton level $E_{\text{sum}} = 1.1716$ (in lattice units) as a Kaluza-Klein resonance at 1.52 TeV. This interpretation is retracted.

Proposition 8 (No Kaluza-Klein interpretation). *The Kaluza-Klein interpretation of the WIN Gravity Theorem is a category error: it imports the assumption that the lattice is a compactified spatial dimension, which contradicts the framework’s central result $\dim \ker D = 4$.*

Proof. The Kaluza-Klein interpretation assumes a compactified spatial dimension of size R and identifies the excited levels with momenta n/R . In HCSM, the lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ is not a compactified spatial dimension: it is the pre-geometric substrate from which spacetime emerges (HCSM-03 [3]). The four-dimensional structure of spacetime is $\ker D$, of dimension 4 (HCSM-02 [2] Proposition 2.1). If the lattice were itself a spatial dimension, spacetime would be five-dimensional rather than emergent, contradicting HCSM-02. $\square$

Remark 9 (The 1.52 TeV number). Independent of the category error, the arithmetic of the WIN-era paper does not close. The naive conversion of $E_{\text{sum}} = 1.1716$ from lattice to physical units gives

$$E_{\text{phys}} = 1.1716 \cdot \frac{M_P^{\text{phys}}}{M_P^{\text{lattice}}} \approx 5.5 \times 10^{13} \text{ TeV}, \quad (9)$$

not 1.52 TeV. Reproducing 1.52 TeV requires a warp suppression $e^{-\alpha k L}$ with $\alpha = 0.812$, which is not a natural exponent. The 1.52 TeV number does not emerge from any HCSM-native conversion.

6 Consequences for gravitational phenomenology

The HCSM framework’s gravitational predictions are now:

- Exactly massless graviton: $m_{\text{graviton}} = 0$ (Theorem 4).
- Standard GR at sub-Planckian scales.
- No fifth force, no scalar graviton.
- No Kaluza-Klein tower.

The absence of a KK tower is a sharp prediction: any detection of a discrete graviton excitation spectrum at accessible energies would falsify the framework.

7 Inputs and derived

7.1 Derived

1. The mass operator structure: any trace-coupled mass operator transforms in A_1 (Proposition 1).
2. The graviton sector: $V_{\text{grav}} = 54B_1 \oplus 54B_2$ lies in $B_1 \oplus B_2$ (Proposition 3).
3. Exact masslessness: $m_{\text{graviton}} = 0$ at all orders (Theorem 4).
4. Absence of finite- N corrections (Proposition 6).
5. Retraction of the finite- N estimate and the KK interpretation.

7.2 Inputs

1. The graviton sector $V_{\text{grav}} = 54B_1 \oplus 54B_2$ (HCSM-45).
2. The D_4 character table and irreps A_1, A_2, B_1, B_2, E (HCSM-05, HCSM-06).
3. The HCSM-00 §9 retirement of the SYK/QIN layer.
4. Schur’s lemma and standard tensor decomposition of a symmetric 2-tensor into trace, traceless-symmetric, and antisymmetric parts (universal mathematics).

7.3 What is not claimed

We do not claim to derive the full nonlinear Einstein equations (subject of HCSM-50). We do not claim to derive the 8π factor in the Newton constant (subject of HCSM-49). We do not claim to derive the graviton self-coupling structure (subject of HCSM-47) or the Faddeev-Popov normalization (subject of HCSM-48). This paper establishes only the exact masslessness of the graviton.

8 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-46_verification.py`. It verifies:

1. The D_4 character table.
2. The orthogonality $\langle A_1, B_1 \oplus B_2 \rangle_{D_4} = 0$ by direct character inner product computation.
3. The graviton polarization character $(2, 2, -2, 0, 0)$.
4. The absence of any D_4 -invariant coupling between A_1 and $B_1 \oplus B_2$ isotypic sectors.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

9 Conclusion

This paper has established the exact masslessness of the graviton in the Hodge Complex Standard Model.

The graviton is the self-paired $B_1 \oplus B_2$ curvature mode of the symmetric tensor square $\text{Sym}^2(\Omega^1|_{\lambda=4})$, with $54B_1 \oplus 54B_2 = 108$ modes. Any trace-coupled mass operator on the $\lambda = 4$ sector transforms in the A_1 isotypic representation of D_4 . Since A_1 and $B_1 \oplus B_2$ are distinct irreps of D_4 , Schur orthogonality gives $m_{\text{graviton}} = 0$ at all orders. The masslessness is exact, not approximate: it holds at all orders in perturbation theory and at all orders in $1/N$, with no finite- N , one-loop, or non-perturbative correction.

The masslessness is the discrete analogue of the continuum statement that diffeomorphism invariance forbids a graviton mass. The exactness of the D_4 symmetry (unlike the approximate symmetry of an effective field theory) makes the masslessness exact.

Two results of the earlier WIN-era Gravity Theorem are retracted. The finite- N graviton mass estimate was an imported effective-field-theory ansatz, and its own arithmetic also disagrees with

the LIGO bound by 13 orders of magnitude. The Kaluza-Klein resonance interpretation of the first excited graviton level imported the assumption that the lattice is a compactified spatial dimension, which contradicts the framework’s central result $\dim \ker D = 4$.

The corrected framework predicts an exactly massless graviton, no fifth force, no scalar graviton, and no Kaluza-Klein tower, all consistent with observation.

References

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A Web Crawler Metadata

A.1 Dublin Core

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A.2 Schema.org JSON-LD

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A.3 BibTeX for Automated Indexing

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           This paper proves that the graviton is exactly massless at all
           orders. Any trace-coupled mass operator transforms in  $A_1$ ; the
           graviton transforms in  $B_1 + B_2$ ; Schur orthogonality gives
            $m_{\text{graviton}} = 0$ . Two WIN-era estimates are retracted.}
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